



Like most great innovations AI is most likely to be co-opted by the military

“Her”, but it’s not too much of a stretch to say that we will be having conversations with our digital assistants in the near future. We wouldn’t go so far as to call Siri and Cortana AI as they are, after all, just algorithms spewing out canned responses and performing a set of pre-defined tasks. They’ve only recently figured out how to relate multiple queries to each other. Siri and Cortana are not Samantha, but we’ve seen their potential. Throw in a chatbot and, who knows, Siri might actually be your next girlfriend.

Again, programs like Siri and Cortana simply interpret sounds, translate it to speech and then use advanced algorithms to determine an appropriate answer. These programs are not thinking, they just do things.

MACHINE LEARNING

How is it that we function? The simple action of walking on two legs is so exceedingly complex for a machine to interpret that we’re yet to see a truly capable “walking robot”. An action that we perform with no conscious thought needs to be defined in its entirety before a machine understands. Such concepts as basic physics, centre of gravity, weight distribution, muscle control, etc. are all hard-wired into our brains. Millions upon millions of years of evolution has automated basic functions, or at least, functions that we perceive as basic. Software can’t do that yet. It can’t take the real world for granted because it is yet to even understand that the real world exists.

Chess is easy. It’s a game with a simple set of rules and a countably finite number

of moves. But the rules of the real world? Even we don’t understand them yet, how do we explain them to a computer? It’s like trying to explain the colour red to a blind man, he will never understand if he’s never seen the colour before. You can explain the physics of colours and light (wavelengths, quanta, reflection, refraction, etc.), and he’ll probably understand the concepts, but he won’t know what the quality of being red really means.

This is where machine learning comes in. A bird is not a physicist, yet it deals with the intricacies of flight on a daily basis and yet, we’ve only now managed to make swarming drones that don’t bump into each other and have figured out how to build a rope bridge only as of last month, that too with the assistance of a supercomputer of sorts.



Pretty decent, but not quite Samantha yet

Obviously, the bird understood the concept of flight by watching its peers and trying for itself (most of the data is still hard-wired of course). Machine Learning works on the same principle. The program is constrained by a certain set of rules and must accomplish certain goals by means of observation and trial and error.

This “learning” AI is proving to be exceedingly popular. A product of Google’s DeepMind project, the Deep-Q Network (DQN) program taught itself to play over 49 Atari 2600 games by simply giving it the goal of achieving the highest possible score. Nobody programmed the rules of the game, nor even the control scheme. The program used a technique called “deep reinforcement learning” to get through the games by trial and error and incessant repetition till it ultimately beat the game.

How long did this take? Just weeks. But it doesn’t stop there. The program started figuring out exploits to take advantage of efficient winning strategies and more. But this is still not true intelligence. Concepts learned in one game are not transferred to the next, it failed miserably at games requiring even a modicum of strategy. Most importantly, despite the program’s skill, it wasn’t even aware of its own existence, let alone the fact that it was playing a game.

As humans, that much intelligence (or lack thereof) is sufficient. Such a program, if trained to drive cars, pilot aircraft or make weather predictions, would just make our lives easier without ever trying to take over the world.

PATTERN RECOGNITION

Another form of intelligence that some might perceive as AI is pattern recognition. Google for example, stores a profile of what it thinks you are based on your browsing habits, Amazon interprets your shopping patterns and can make intelligent suggestions and YouTube music knows your taste in music.

All of this is the result of data mining. Services such as Google and Amazon harvest vast amounts of data about you. Given enough data, patterns begin to emerge. The data collected is so vast and so thorough that usually the profile that they can create to describe you (based on collected data) will be better than anything that even your psychiatrist would